

AAKRITI KUMAR

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CURRENT APPOINTMENT

Postdoctoral Researcher

Aug 2024-

Kellogg School of Management, Northwestern University
Northwestern Institute on Complex Systems

EDUCATION

PhD in Cognitive Science

2018 - 2024

University of California, Irvine

Thesis: Human Mental Models of Self, Others, and AI Agents

MS in Statistics

2018 - 2022

University of California, Irvine

BTech in Engineering

2011 - 2015

Indian Institute of Technology, Madras

RESEARCH EXPERTISE

Human-AI Collaboration, Computational Social Science, Behavioral Science, Cognitive Science, Statistical Modeling, Team Dynamics, Digital Experimentation

PEER-REVIEWED PUBLICATIONS

- [1] **A. Kumar**, N. Poungpeth, D. Yang, E. Farrell, B. Lambert, M. Groh (2026). [When Large Language Models are Reliable for Judging Empathic Communication](#). *Nature Machine Intelligence*
- [2] **A. Kumar**, R. Tham, M. Steyvers (2025). [Seeing Things Differently: The Role of Differing Perspectives in Advice-Taking](#). *Cognitive Science (Vol. 47)*
- [3] M. Steyvers, H. Tejada, **A. Kumar**, S. Karny, X. Hu, L. Mayer, P. Smyth (2025). [What large language models know and what people think they know](#). *Nature Machine Intelligence*
- [4] N. Kamali, K. Nakamura, **A. Kumar**, A. Chatzimparpas, J. Hullman, M. Groh (2025). [Characterizing Photorealism and Artifacts in Diffusion Model-Generated Images](#). *CHI 2025*
- [5] **A. Kumar**, P. Smyth, M. Steyvers (2023). [Differentiating mental models of self and others: A hierarchical framework for knowledge assessment](#). *Psychological Review*
- [6] M. Steyvers & **A. Kumar**, (2023). [Three Challenges for AI-Assisted Decision-Making](#). *Perspectives on Psychological Science*
- [7] M. Kelly, **A. Kumar**, P. Smyth, M. Steyvers (2023). [Capturing Humans' Mental Models of AI: An Item Response Theory Approach](#). *FAccT 2023*
- [8] **A. Kumar**, K. Akash, S. Mehrotra, T. Misu, M. Steyvers (2023). [When Do Drivers Intervene In Autonomous Driving?](#). *Human Robot Interaction (Late-Breaking Report) 2023*
- [9] H. Lemus, **A. Kumar**, M. Steyvers (2022). [How Displaying AI Confidence Affects Reliance and Hybrid Human-AI Performance](#). *Proceedings of the Second International Conference on Hybrid Human-Machine Intelligence*.
- [10] H. Lemus, **A. Kumar**, P. Smyth, M. Steyvers (2022). [AI-assisted Decision-Making: A Cognitive Modeling Approach to Infer Latent Reliance Strategies](#). *Computational Brain & Behavior*

- [11] H. Lemus, **A. Kumar**, M. Steyvers (2022). [An Empirical Investigation of Reliance on AI-Assistance in a Noisy-Image Classification Task](#). *Proceedings of the First International Conference on Hybrid Human-Machine Intelligence*.
- [12] **A. Kumar***, S. Chatterjee¹, P. Shenoy (2022). [Meta-Learning of Dynamic Policy Adjustments in Inhibitory Control Tasks](#). *Cognitive Science (Vol. 44)*.
- [13] **A. Kumar**, T. Patel, A. Benjamin, M. Steyvers (2021). [Explaining Algorithm Aversion with Metacognitive Bandits](#). *Cognitive Science (Vol. 43)*
- [14] **A. Kumar**, A. Benjamin, A. Heathcote, M. Steyvers (2021). [Comparing models of learning and relearning in large-scale cognitive training data sets](#). *NPJ Science of Learning*.
- [15] J. Rouder, **A. Kumar**, J. Haaf (2019). [Why Most Studies of Individual Differences With Inhibition Tasks Are Bound To Fail](#). *Psych Bulletin & Review*

PREPRINTS & DRAFTS

- [P1] **A. Kumar**, F. Pongpeth, D. Yang, B. Lambert, M. Groh. [Practicing with Language Models Cultivates Human Empathic Communication](#). *arXiv*
- [P2] N. Dundas*, **A. Kumar***, Z. Guo, O. Badri, A. Koochek, M. Groh. Mapping Diagnostic Blind spots by Physicians and AI in Dermatology.
- [P3] N. Kamali, **A. Kumar**, A. Chatzimpampas, C. Gerstner, J. Hullman, M. Groh. Guiding Human Visual Attention to Spot AI-Generated Images.
- [P4] Z. Guo, **A. Kumar**, S. Melumad, Z. Bucinca, M. Groh. Metacognitive Forcing Functions to Preserve Human Learning Depth in LLM Interactions.
- [P5] **A. Kumar**, P. Petrides, K. Chauncey. Simulating AV System Delays to Study User Stress & Behavioral Intention.

PRESENTATIONS AND TALKS

Invited Talks

1. AoM Symposium: Augmenting Human Potential with Generative AI: Opportunities and Risks. *Organizer, AoM 2026*
2. Building custom tools for studying social behavior. *Invited speaker, PDW on GenAI as a Window into Organizations: A Window into Cognitive and Behavior, AoM 2026*
3. Cultivating Human Empathy with AI Role-playing Partners. *Mind and Machine Alignment Summit, Ohio State University, July 2026*
4. Cultivating Human Empathy with AI Role-playing Partners. *AI and Social Research: Empathic AI, Metascience, and Methodology, Penn State University, June 2026*
5. Enabling Physician-AI Collaboration by Mapping Diagnostic Blind spots & Edges. *MORS Brownbag, May 2026*
6. Practicing with Language Models Cultivates Human Empathic Communication. *Guest Speaker, COG SCI 366: Cognitive Science Research Seminar, Northwestern University, April 2026*
7. Cultivating Human Empathy with AI Partners. *Invited Speaker, DySoc Seminar 2025, University of Tennessee, Knoxville*
8. Cultivating Human Empathy with AI Partners. *Guest Speaker, Learn Behavioral Leadership Symposium, Chicago*
9. Large language models can provide expert-aligned judgments of empathic communication. *Lightning Talk, Northwestern Institute on Complex Systems Seminar, 2025*
10. What Decision-Makers Should Know About Generative AI. *JTIP Symposium 2025: Innovation, AI, and Access, Northwestern School of Law*

¹* indicates equal contribution

11. AI-assisted Decision Making. *UC Noyce Initiative Symposium, 2022*
12. Understanding Human-AI Teams. *Noyce Symposium, 2022*
13. Member, Panel on Human-AI Relationships. *Noyce Symposium, 2022*

Conference Talks

1. Mapping the Idioms of Empathy: A Data-Driven Taxonomy of Empathic Communication Using k-Sparse Autoencoders. *12th International Conference on Computational Social Science (IC2S2), Burlington, VT, July 2026*
2. Practicing with Language Models Cultivates Human Empathic Communication. *12th International Conference on Computational Social Science (IC2S2), Burlington, VT, July 2026*
3. Cultivating Human Empathy with AI Role-playing Partners. *Parallel Talk, Conference on Digital Experimentation @ MIT, 2025*
4. Seeing Things Differently: The Role of Differing Perspectives in Advice-Taking. *Oral Talk, 47th Annual Meeting of the Cognitive Science Society, San Francisco, August 2025*
5. Metacognitive Bandits: When Do Humans Seek AI Assistance? *Social Intelligence in Humans and Robots Workshop, ICRA 2021*
6. Explaining Algorithm Aversion with Metacognitive Bandits. *Spotlight talk, Workshop on Human-AI Collaboration in Sequential Decision-Making, ICML 2021*
7. Explaining Algorithm Aversion with Metacognitive Bandits. *Oral Talk, 43rd Annual Meeting of the Cognitive Science Society, August 2021*

Conference Posters

1. Practicing with Language Models Cultivates Human Empathic Communication. *Symposium on Communication & Intelligence, University of Chicago, Chicago, IL, May 2026*
2. Practicing with Language Models Cultivates Human Empathic Communication. *Wharton AI & the Future of Work Conference, May 2026*
3. When Large Language Models are Reliable for Judging Empathic Communication. *International Conference on Computational Social Science (IC2S2), Norrkoping, Sweden, July 2025*
4. Evaluating Elements of Empathic Communication with Experts, Crowds, and Large Language Model. *International Conference on Computational Social Science (IC2S2), Norrkoping, Sweden, July 2025*
5. Evaluating Human Perception of AI-Generated Images Across Pose Complexities. *International Conference on Computational Social Science (IC2S2), Philadelphia, PA, July 2024*
6. An Empirical Investigation of Reliance on AI-Assistance in a Noisy-Image Classification Task. *Workshop on Trust and Reliance in AI-Human Teams, CHI, April 2023*

AWARDS AND FELLOWSHIPS

UCI Associate Dean's Fellowship 2023, UCI	<i>Total Award: \$7000</i>
Noyce Initiative in AI, Law, and Society Fellowship 2021-2023	<i>Total Award: \$66,000</i>
Graduate Dean's Recruitment Fellowship 2018, UCI	<i>Total Award: \$5000</i>
Recruitment Fellowship Award, 2018, Department of Cognitive Science, UCI	<i>Total Award: \$5000</i>

INDUSTRY EXPERIENCE

Quantitative Human-Computer Interaction Intern, Motional	<i>Apr'23 - Dec'23</i>
<i>Human Assistance in Self-Driving Robotaxis:</i> Investigated Motional's remote vehicle assistance capability in assisting their robotaxi. [P5]	
Human Behavior Modeling Intern, Honda Research Institute	<i>Jun'22 - Sept'22</i>
<i>Trust in Autonomous Vehicles:</i> Created a quantitative model to capture the effect of an autonomous vehicle's driving style on users' trust and risk perception. Identified scenarios perceived as risky across two mobility types [8]	

Statistical Consultant, Google Research, India*Jan'20 - Jan'21*

AI for Social Good: Helped design data collection and analysis methods to evaluate an ML solution's efficacy in reducing drop-off from an Indian maternal health program. *Reliability in People's Inhibitory Control*: Proposed and tested RNN models to assess people's cognitive control task traits are reliable. [12]

TEACHING EXPERIENCE

Teaching Assistant, UC Irvine: Probability and Statistics in Psychology I, II, III *Fall 2019 - Spring 2020*

Teaching Assistant, UC Irvine: Creativity *Spring 2019*

Lead Teaching Assistant, UC Irvine: Introduction to Human Memory *Winter 2019*

Teaching Assistant, UC Irvine: Introduction to Psychology *Fall 2018*

MEMBERSHIPS AND SERVICE

Academy of Management Member

Judge, CoDEX 2026 Poster Session, Northwestern University *Apr 2026*

Reviewer (Academy of Management, Transactions on Interactive Intelligent Systems, Trends in Cognitive Sciences, Cognitive Science, Human-Robot Interaction, CHI, Applied Psychology: Health and Well-Being, Judgment and Decision-making) *2020 - Present*

Department Colloquium Coordination Committee Member *2021 - 2022*

Cognitive Science Society Student Member